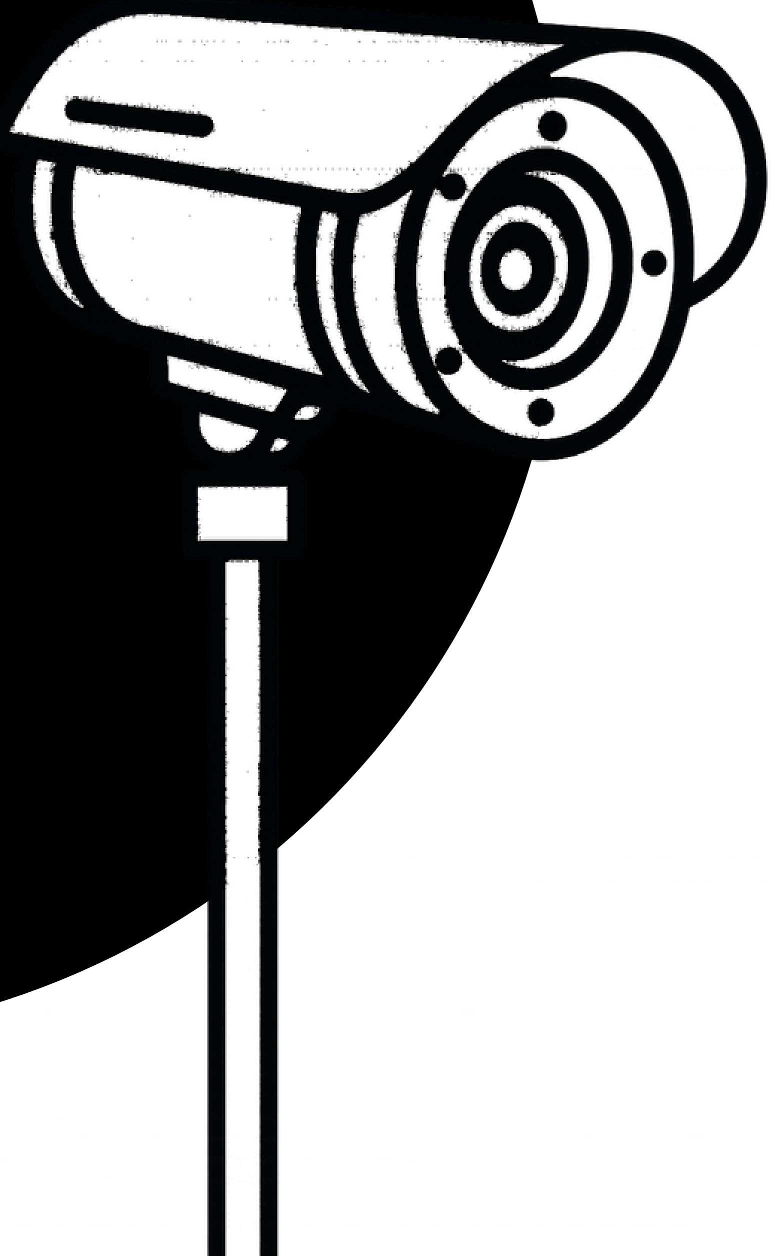
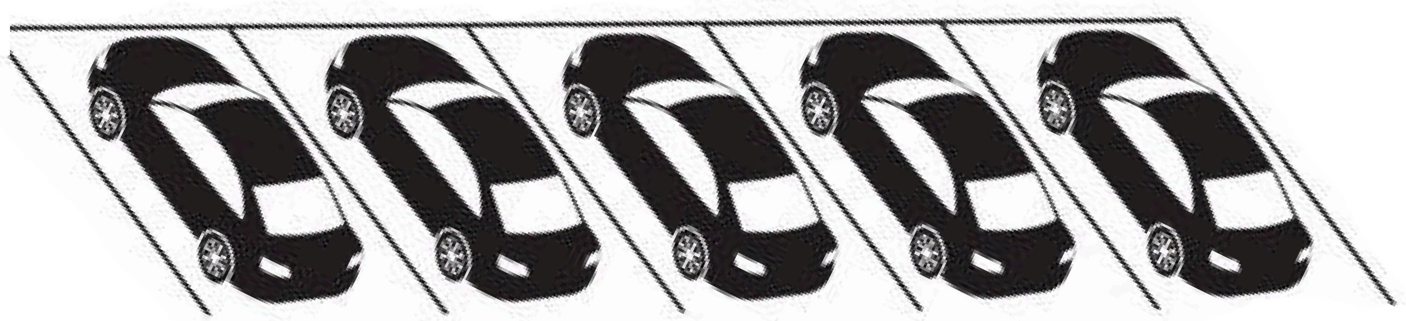
GITHUB

ATU FINAL YEAR PROJECT
SEAN MALONEY

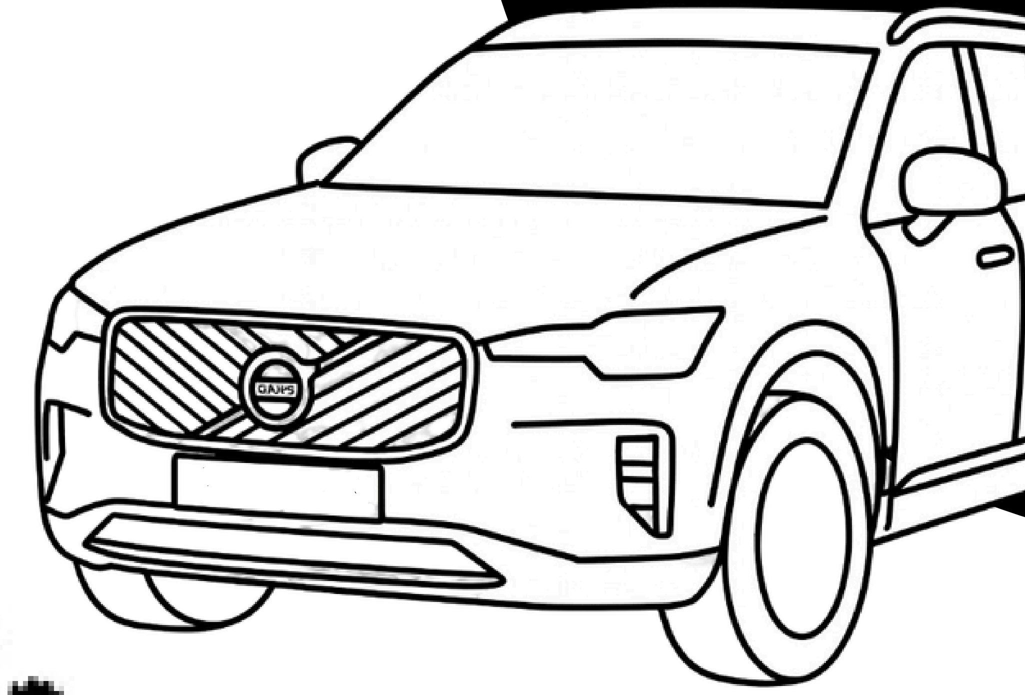
PARKING FINDER



SUMMER DEMO

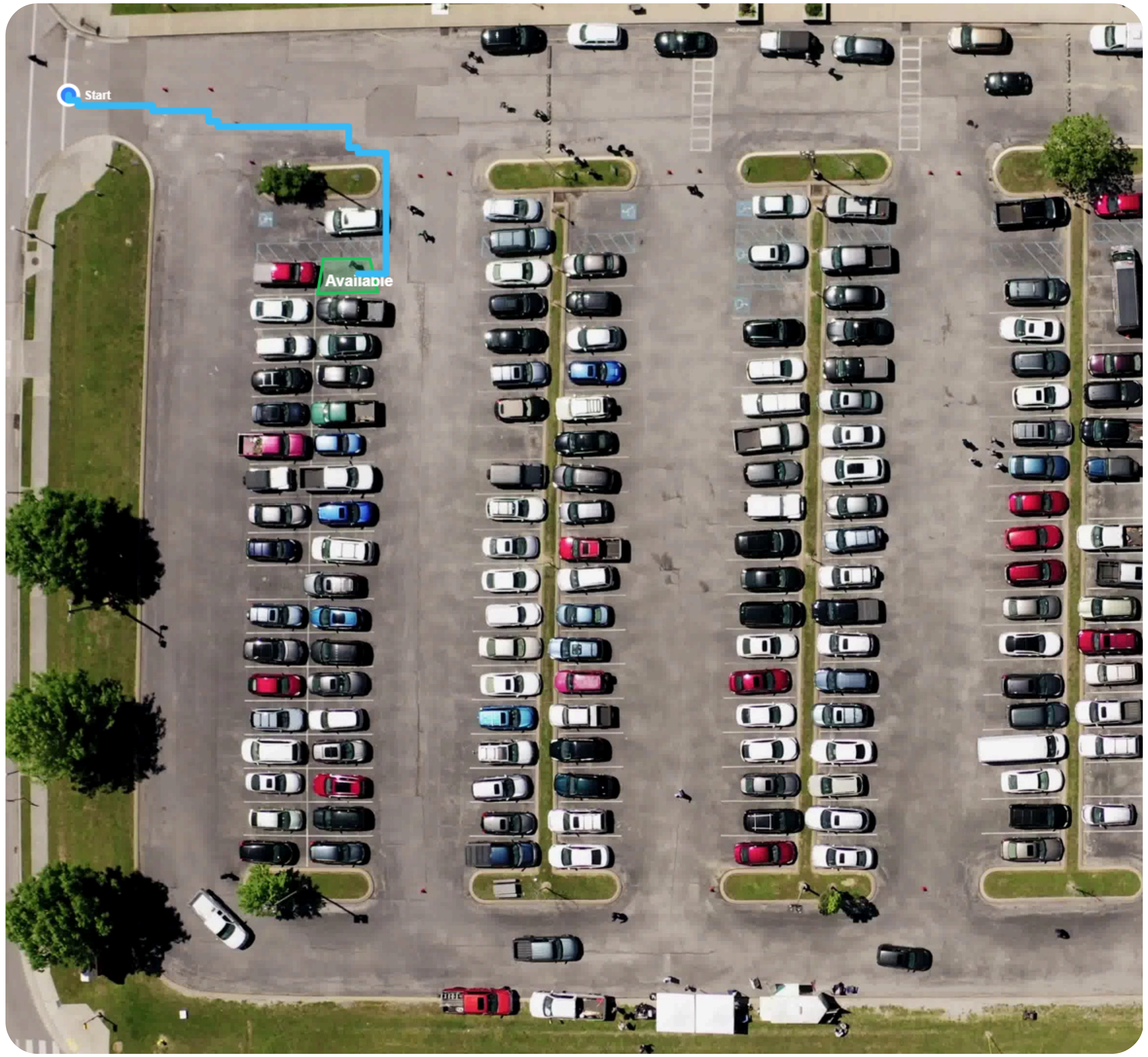


The most efficient problem for tackling modern problems is using modern solutions.

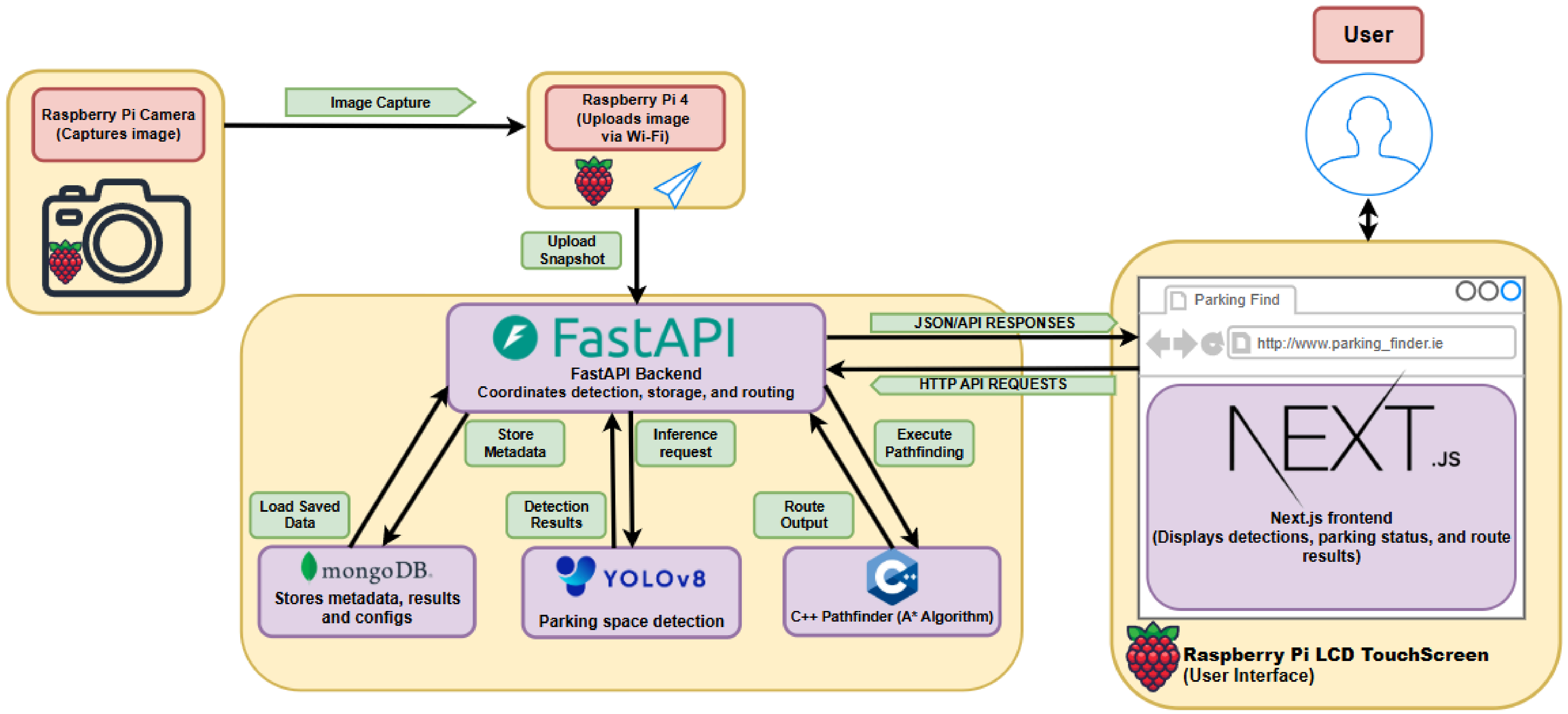


THE PROJECT OVERVIEW

- Smart parking prototype using AI and computer vision
- Detects vehicles and classifies spaces as free or occupied
- Built with YOLOv8, Raspberry Pi, FastAPI, MongoDB, and Next.js
- Includes parking visualisation and A* route generation
- Designed as a low-cost parking assistance solution



FINAL ARCHITECTURE DIAGRAM



SYSTEM OVERVIEW

01

Parking Finder is a smart parking system that integrates image capture, AI detection, parking analysis, and route generation.

IMAGE CAPTURE & UPLOAD

02

A Raspberry Pi camera captures parking images, and the Raspberry Pi 4 uploads them to the backend over Wi-Fi.

BACKEND, AI & ROUTING

03

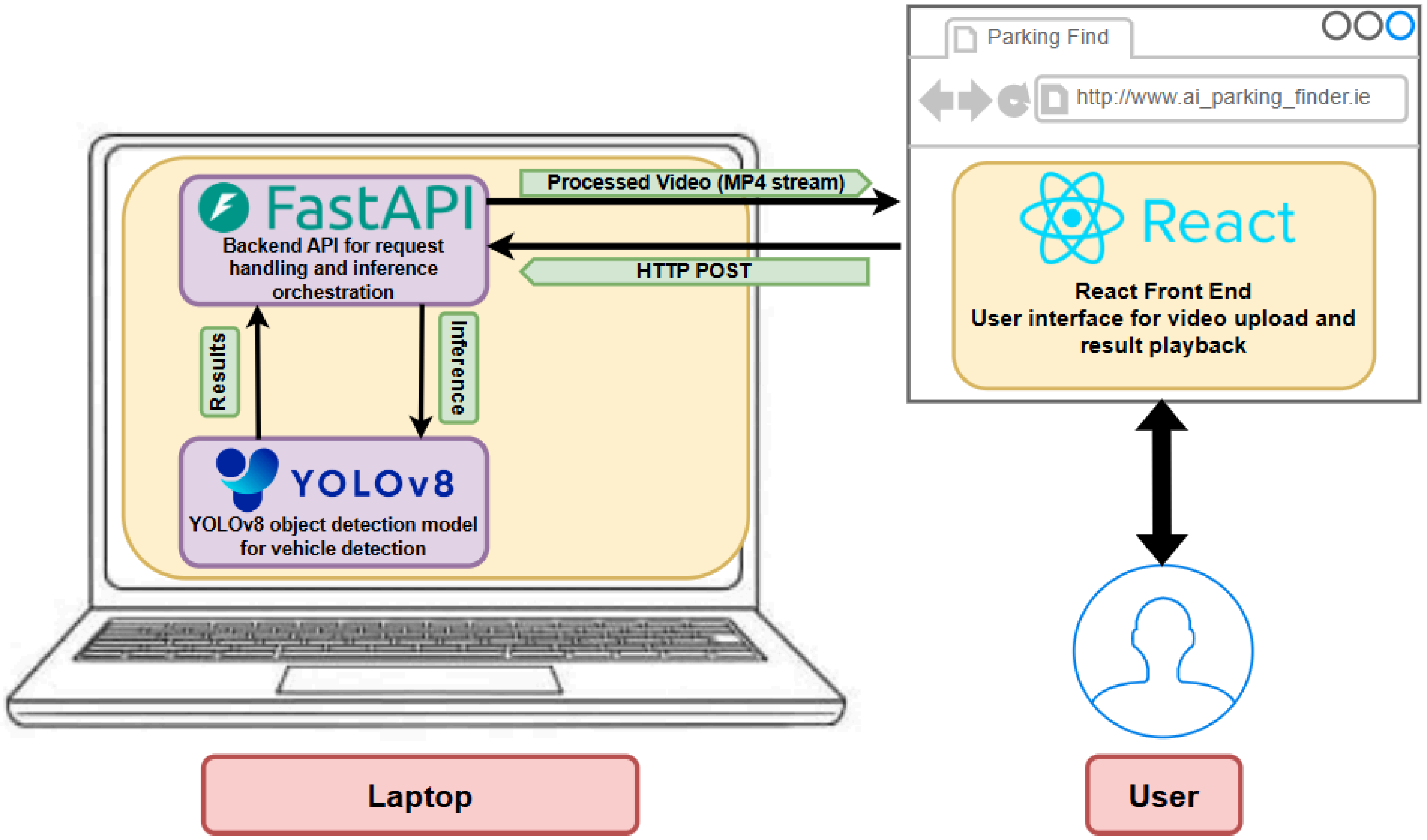
FastAPI manages the core logic, connecting MongoDB storage, YOLOv8 vehicle detection, and C++ A* pathfinding.

USER INTERACTION & OUTPUT

04

A Next.js frontend on a Raspberry Pi LCD touchscreen displays parking status, detections, and route results to the user.

WHAT'S NEW SINCE CHRISTMAS DEMO?



01 PI INTEGRATION

Added Raspberry Pi camera capture, Raspberry Pi 4 upload flow, and LCD touchscreen support.

02 DATABASE & STORAGE

Integrated MongoDB for persistent storage of parking configurations, metadata, and capture data.

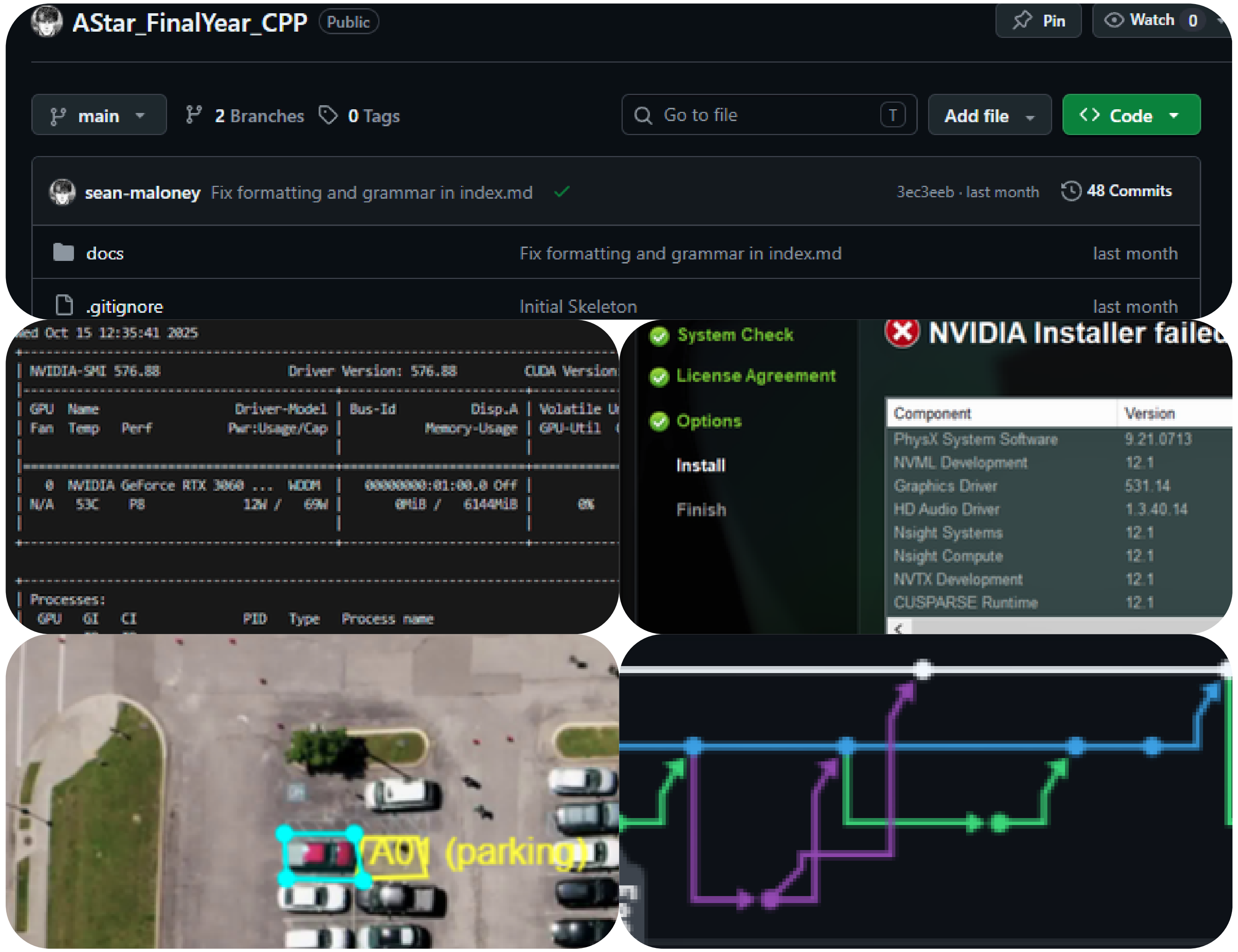
03 PARKING ANALYSIS & ROUTING

Added parking polygon configuration, occupancy classification, and C++ A* pathfinding to generate routes to available spaces.

04 FRONTEND & SYSTEM IMPROVEMENTS

Redesigned the frontend with an Ookla Speedtest inspired layout and improved overall system integration across hardware, backend, AI, and routing.

CHALLANGES AND SOLLUTIONS



01

GPU TRAINING & CUDA COMPATIBILITY

Challenge: Training was initially on CPU instead of GPU due to CUDA, PyTorch, and driver version mismatches.

Solution: Researched CUDA compatibility matrices, aligned software versions to successfully enable GPU.

02

CAMERA APPROACH & SYSTEM DIRECTRION

Challenge: The original drone based idea was not practical for a realistic parking deployment.

Solution: Redesigned around a fixed elevated camera for a more practical and reliable solution.

03

OCCUPANCY CLASSIFICATION

Challenge: Vehicle detection alone could not identify free spaces.

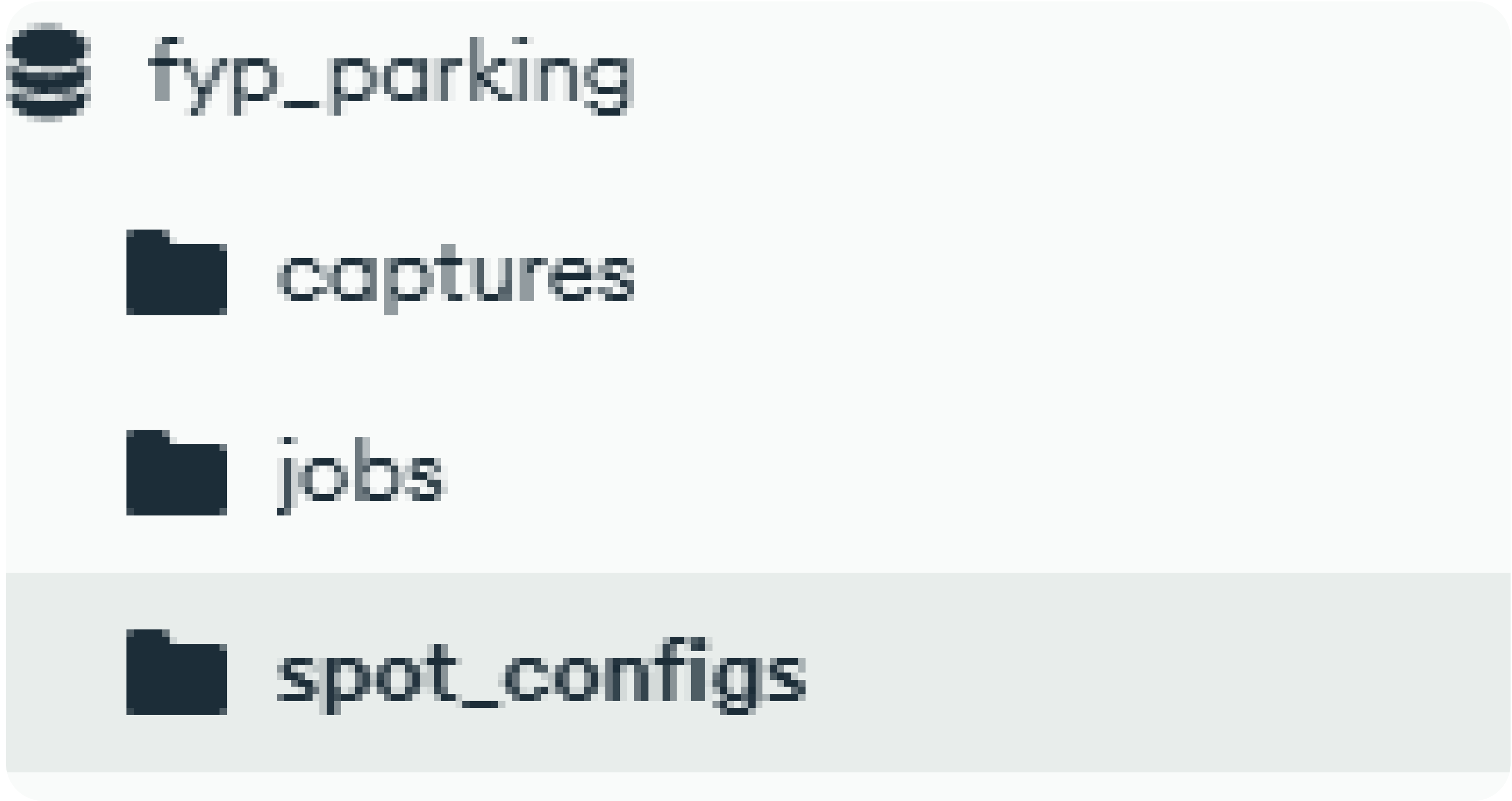
Solution: Parking polygons were added for space classification.

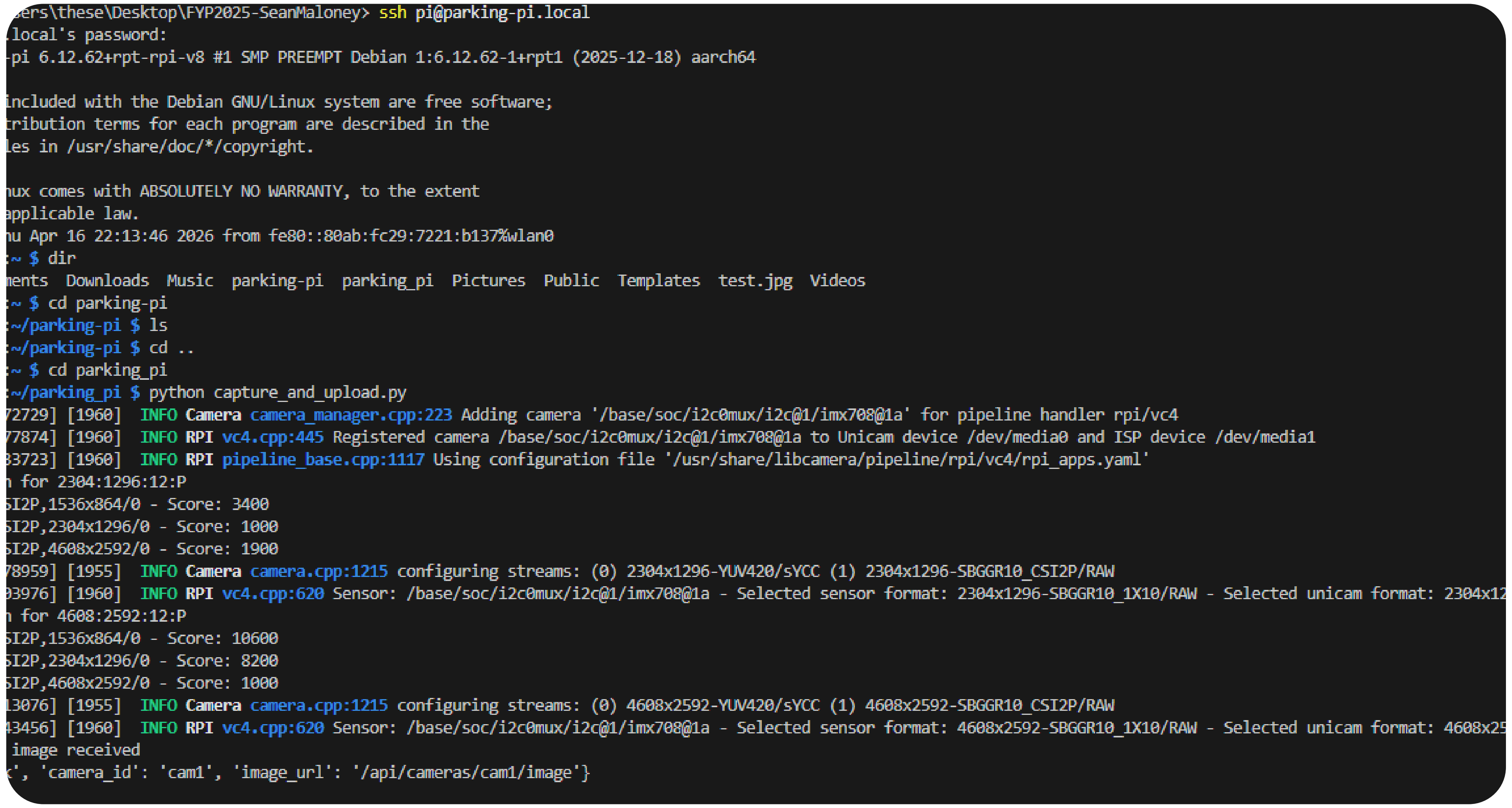
04

PATHFINDING & GRID SETUP

Challenge: Route generation depended on accurate grid mapping.

Solution: A C++ A* system was integrated for routing.

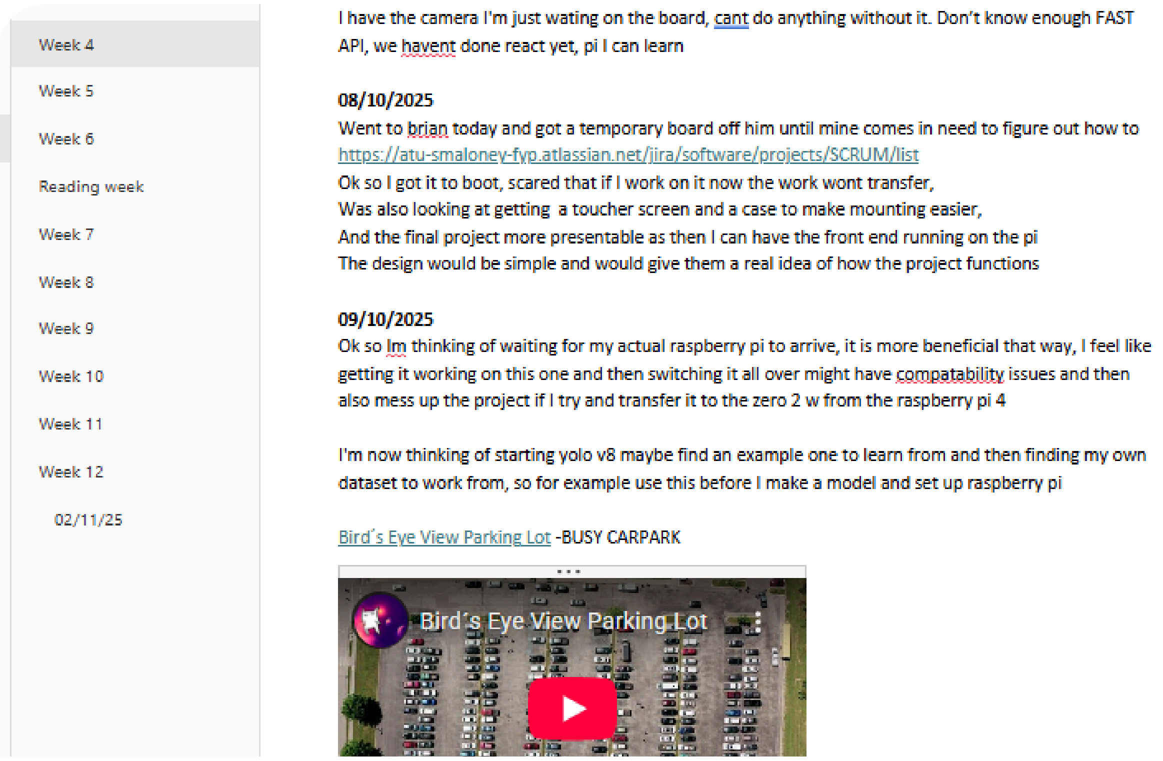




RESEARCH

- Parking space configuration
- Occupied vs free space detection
- A* route generation
- Grid layout mapping
- MongoDB data storage
- Raspberry Pi integration

PROJECT MANAGEMENT 1



Week 4
Week 5
Week 6
Reading week
Week 7
Week 8
Week 9
Week 10
Week 11
Week 12
02/11/25

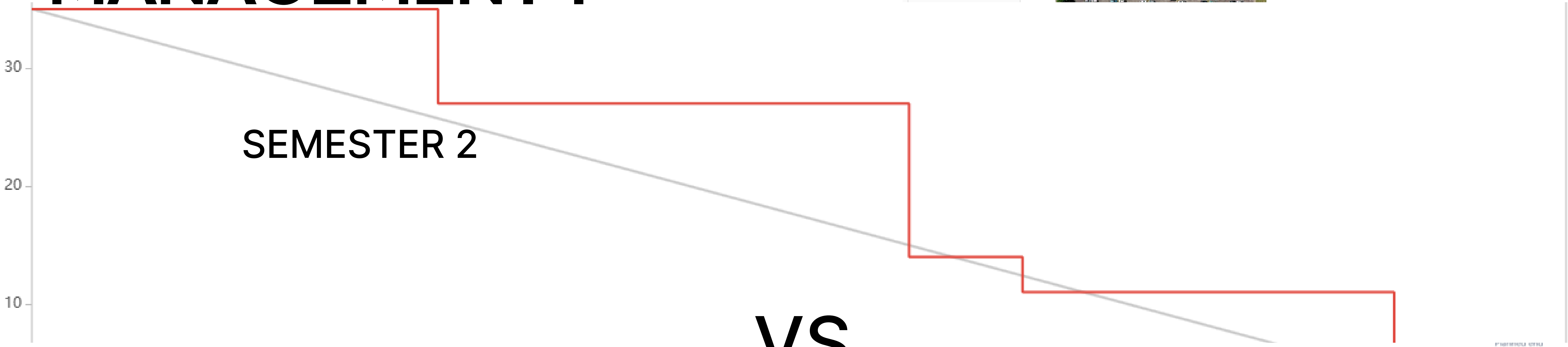
I have the camera I'm just wating on the board, cant do anything without it. Don't know enough FAST API, we havent done react yet, pi I can learn

08/10/2025
Went to brian today and got a temporary board off him until mine comes in need to figure out how to
https://atu-smaloney-fyp.atlassian.net/jira/software/projects/SCRUM/list
Ok so I got it to boot, scared that if I work on it now the work wont transfer,
Was also looking at getting a toucher screen and a case to make mounting easier,
And the final project more presentable as then I can have the front end running on the pi
The design would be simple and would give them a real idea of how the project functions

09/10/2025
Ok so Im thinking of waiting for my actual raspberry pi to arrive, it is more beneficial that way, I feel like getting it working on this one and then switching it all over might have compatability issues and then also mess up the project if I try and transfer it to the zero 2 w from the raspberry pi 4

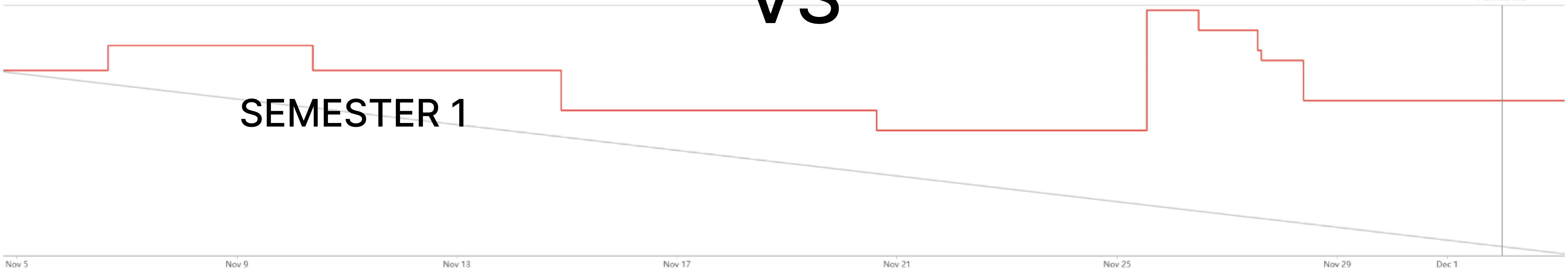
I'm now thinking of starting yolo v8 maybe find an example one to learn from and then finding my own dataset to work from, so for example use this before I make a model and set up raspberry pi

Bird´s Eye View Parking Lot -BUSY CARPARK



SEMESTER 2

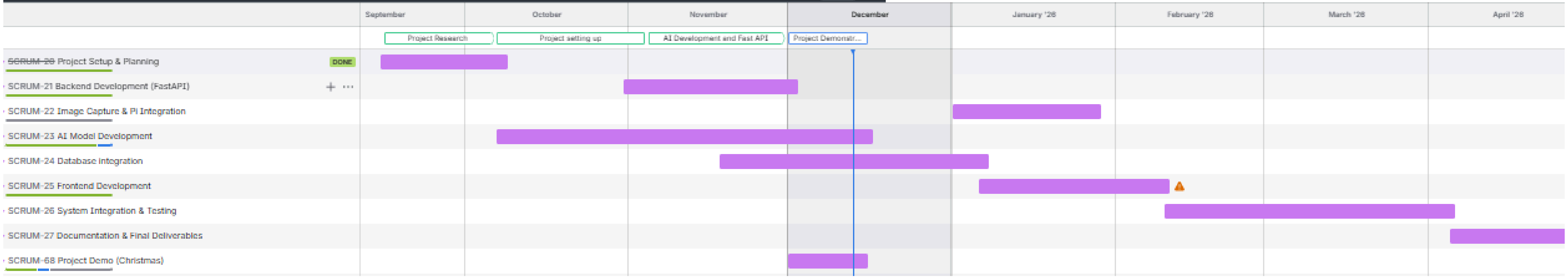
VS



SEMESTER 1

PROJECT MANAGEMENT 2

- Sprint based development
- Clear improvement from Semester 1→ Semester 2
- Better task breakdown and estimation over time
- Moved to 2/1 week sprints for easier management
- Gannt chart used for easier planning.
- Jira used to track sprint progress and priorities
- GitHub network reflects improving and structured workflow
- OneNote used to document project progress



YOLO AND A*

01

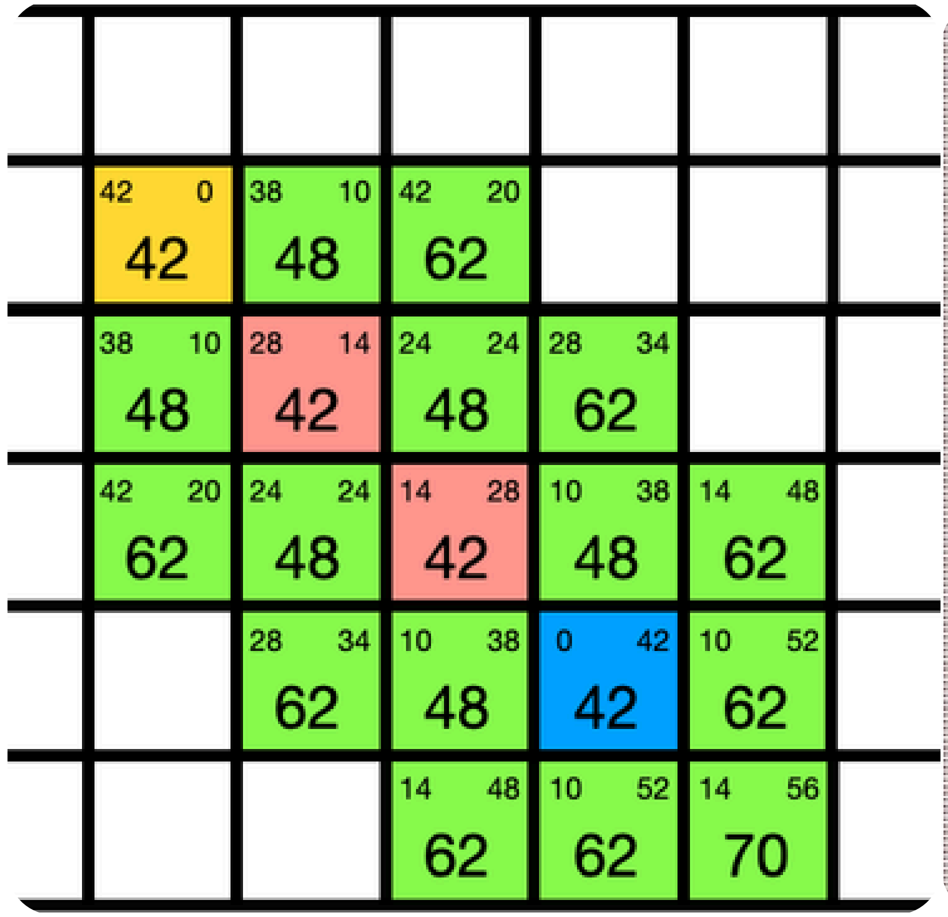
C++ A* Algorithm Pathfinder

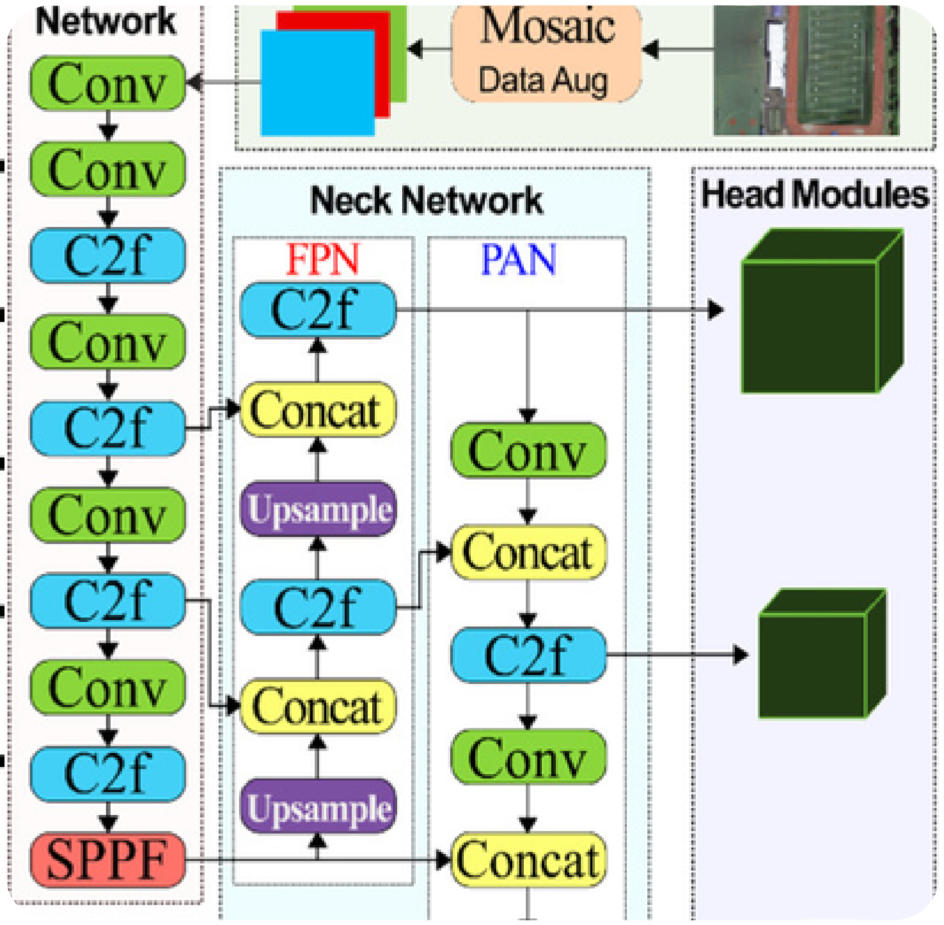
A* is a pathfinding algorithm used to find the shortest or most efficient path between two points.
It works by comparing the cost already travelled with an estimate of the cost remaining to the goal.

02

YOLOv8 Vehicle Detection

YOLOv8 is an object detection model used to identify and locate objects within an image.
It works by predicting bounding boxes and class labels in a single pass, making it fast and effective for real time detection.

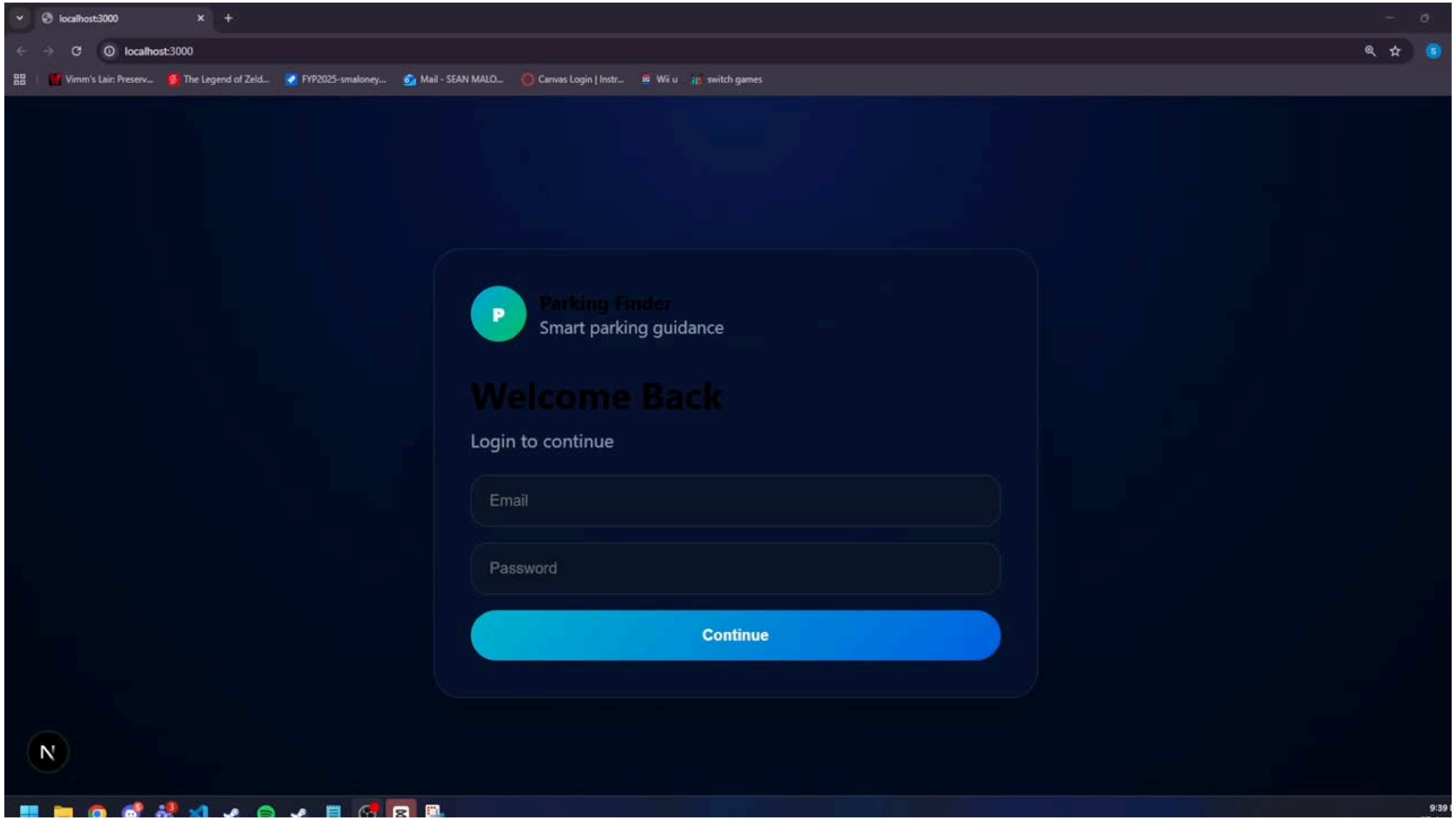




AI DECLERATION

·Generative AI tools were used as support tools during the preparation of this project.
·AI assistance, where used, was used for tasks such as brainstorming, language refinement, formatting support, structure planning, debugging, generation of code, rewriting of code, and web scraping information not easily found.
·All technical design, implementation, testing, evaluation, and final written content were reviewed, validated, and approved by the author.
·The author takes full responsibility for the accuracy, originality, and academic integrity of this submission.

DEMO VIDEO


localhost:3000
localhost:3000
P
Parking Finder
Smart parking guidance
Welcome Back
Login to continue
Email
Password
Continue